

# Full-Stack + AI/ML Developer Roadmap — 45 Projects from Beginner to Job-Ready

A comprehensive guide to becoming a professional Full-Stack and Machine Learning Developer using the MERN Stack and TensorFlow/Keras.

# Section 1: Introduction

This document is a structured roadmap containing 45 projects, starting from beginner to advanced, designed for developers aspiring to become Full-Stack + AI/ML professionals. It includes the tools, frameworks, and step-by-step logic for each project, along with deployment strategies and portfolio tips.

## Core Technologies:

- 1 Frontend: React.js, TailwindCSS, Material UI
- 2 Backend: Node.js, Express.js, MongoDB
- 3 Machine Learning: TensorFlow, Keras, Scikit-learn (for preprocessing)
- 4 Version Control: Git, GitHub
- 5 Hosting: Vercel, Netlify, Render, MongoDB Atlas

## Phase 1 — Beginner Foundations (HTML, CSS, JS, React Basics)

Focus on front-end skills and web fundamentals.

## Phase 2 — Intermediate Projects (Full-Stack & React Proficiency)

Build scalable full-stack web apps using the MERN stack.

## Phase 3 — Machine Learning & AI Integration (TensorFlow Focus)

Integrate ML models into web apps using TensorFlow and Python APIs.

## Phase 4 — Advanced Full-Stack + AI Systems

Develop robust, production-grade apps combining full-stack and AI features.

## Phase 5 — Capstone & Professional SaaS Projects

Create portfolio-ready or startup-level AI-driven full-stack applications.

### ■ Personal Portfolio Website

Showcase your projects and resume.

**Tech Stack:** React, TailwindCSS, GitHub Pages

**Skills Learned:** Frontend foundations, hosting, design.

### ■ Task Manager App

Full CRUD task system.

**Tech Stack:** React, Node.js, Express, MongoDB

**Skills Learned:** Full-stack fundamentals.

### ■ Movie Recommendation System

Suggests movies using collaborative filtering.

**Tech Stack:** Python, TensorFlow, React

**Skills Learned:** ML integration.

## ■ Chatbot Assistant

AI-powered chatbot using OpenAI API.

**Tech Stack:** React, Node.js, TensorFlow

**Skills Learned:** NLP, API integration.

## ■ Stock Market Dashboard

Real-time stock tracking and prediction.

**Tech Stack:** React, WebSockets, TensorFlow

**Skills Learned:** Real-time data, ML predictions.

## Section 3: Frameworks and Tools Reference

- 1 Frontend: React.js (<https://react.dev>), TailwindCSS (<https://tailwindcss.com>)
- 2 Backend: Node.js (<https://nodejs.org>), Express.js (<https://expressjs.com>)
- 3 Database: MongoDB (<https://www.mongodb.com/atlas>)
- 4 Machine Learning: TensorFlow (<https://www.tensorflow.org>), Keras, Scikit-learn
- 5 Data Visualization: Chart.js, Plotly, D3.js
- 6 APIs: OpenAI API, Weather API, NewsAPI, Stripe, Twilio
- 7 Hosting: Vercel, Netlify, Render, MongoDB Atlas

## Section 4: Deployment & Portfolio Strategy

- Use GitHub for version control and commit regularly. - Host frontend apps on Vercel or Netlify. - Deploy backend APIs to Render or Railway. - Store data securely using MongoDB Atlas. - Link your deployed apps to your portfolio site. - Document every project with a clear README including tech stack, features, and demo link.

## Section 5: Final Career Roadmap

1. Master MERN fundamentals (Frontend + Backend). 2. Learn Python and TensorFlow for applied AI/ML. 3. Integrate ML into full-stack apps. 4. Build 2–3 large-scale portfolio projects. 5. Contribute to open-source or freelance projects. 6. Apply for full-stack or ML developer roles with a polished portfolio.